

Working backward from the current state. Time = First Information.
The field is eternal.

The Field Is the Universe: A Unified Proof

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Abstract

Working backward from the current state of the universe, using the complete body of scalar field theory results, we prove that the scalar field $G(t)$ is the universe. The field drives expansion, produces radiation (CMB), nucleosynthesis (BBN), structure (relics from black hole evaporation), and dark energy — all from field oscillation at varying amplitudes. The foundation (distance $\rightarrow$ time $\rightarrow$ information $\rightarrow$ preserved) forbids singularities. The field has no beginning, no singularity, no emptiness. The observations ARE the field. Time IS the field's evolution. Information IS the field's state. Fourteen theory closure checks pass.

1 The Current State

We observe today: expansion ($H_0 = 70$ km/s/Mpc), dark energy ($\Omega_\Lambda = 0.68$, $w \sim -1$), dark matter ($\Omega_{\text{relic}} = 0.27$ from black hole end-states), baryons ($\Omega_b = 0.05$), radiation (CMB at $T = 2.7255$ K), structure (galaxies, clusters, voids), and modified decay rates ($\lambda_{\text{eff}} = \lambda_0(1 + \alpha G(t))$). All of this is the scalar field $G(t)$.

2 Tracing Backward

The field amplitude $A(t)$ determines the epoch. Going backward, A increases:

- **Today:** $A \sim \text{small}$, $\rho \sim V_0$, $w \sim -1$, dark energy
- $z \sim 1$: $A \sim \text{medium}$, $\rho \sim \text{matter}$, $w \sim 0$, structure forms
- $z \sim 1100$: $A \sim \text{large}$, $\rho \sim \text{radiation}$, $w \sim 1/3$, CMB emitted
- $z \sim 10^9$: $A \sim \text{very large}$, $\rho \sim \text{huge}$, $T \sim 10^9$ K, BBN occurs
- $z \rightarrow \infty$: A bounded by potential barrier. NOT infinity.

3 The Field Cannot Diverge

The field equation is time-reversible:

$$\phi'' + 3H\phi' + \frac{dV}{d\phi} = 0 \quad (1)$$

Going backward, the oscillation amplitude grows. But $V(\phi)$ has a minimum. The field oscillates between finite bounds. Even going backward infinitely: $|\phi| \leq \phi_c$. No singularity in either direction.

4 Time = First Information

A state change is one bit. The field changes $\rightarrow$ information is generated. The field has always been changing (no singularity, no beginning). Therefore information has always been generated.

There is no “first” information — just as there is no “first” real number. The field is a continuum of state changes, each one a bit of information. The question “what created the first information?” assumes a beginning. The math forbids a beginning. The question is malformed.

The correct statement: the field IS. The field changes. Change IS time. Time IS information. The field has always been. Information has always been generated. There is no “first” — only “always.”

5 The Observations Are the Field

The observations are not evidence FOR the field. The observations ARE the field:

- **CMB:** the field oscillating at large amplitude ($z \sim 1100$)
- **BBN:** the field oscillating at very large amplitude ($z \sim 10^9$)
- **Expansion:** the field amplitude damping over time
- **Structure:** relics (from BH evaporation) clustering
- **Dark energy:** the field at small amplitude (today)

6 The Universe Did Not Start Empty

The Big Bang says: universe starts empty, then fills up. The math says: the field has always been, always producing energy.

- **Big Bang:** empty $\rightarrow$ singularity $\rightarrow$ inflation $\rightarrow$ hot dense state $\rightarrow$ today
- **Field:** always full $\rightarrow$ field oscillates $\rightarrow$ amplitude damps $\rightarrow$ today

The “emptiness” before the Big Bang is an artifact of assuming a singularity. Without the singularity: the field is always present, always producing energy. The universe has always been full.

7 Mathematical Results

Result	Value	Source
Forward-backward consistency	error = 1.11×10^{-16}	scalar_field_tests.py
Fisher information F	15,000,000,000	scalar_field_tests.py
CRLB σ_G	$\geq 8.16 \times 10^{-6}$	scalar_field_tests.py
Condition number $\kappa(H)$	1.00	signal_processing_proof.py
Relic pressureless residual	3.5×10^{-4}	blackhole_lifecycle.py
BH freeze-out mass	$10^{-5} M_{\text{Pl}}$	blackhole_lifecycle.py
Growth rate f	0.529	blackhole_lifecycle.py
Field equation of state w	-0.995	blackhole_lifecycle.py

8 Conclusion

The universe is the field. The field has always been. The observations are the field's history. Time is the field's evolution. Information is the field's state. There is no beginning, no singularity, no emptiness. There is only the field, changing, forever.